

Notes: Gentzkow and Shapiro (QJE, 2011)

Ideological Segregation Online and Offline

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	Online news data	3
3.2	Offline media data	3
3.3	Face-to-face interaction data	4
3.4	Share conservative and conservative exposure	4
3.5	Isolation index	4
3.6	Moderates and ideology imputation	5
4	Models and empirical specifications	5
4.1	Main comparison across media	5
4.2	Leave-out estimation	6
4.3	Distribution of exposure	6
4.4	Time-series specification	6
4.5	Mechanism analysis	6
4.6	Robustness and story-level checks	6
5	Identification and empirical design	7
5.1	What the design can identify	7
5.2	Why the comparison is informative	7
5.3	Main threats and limitations	7
6	Tables and figures	8
6.1	Figure I and Table I: Measurement validation and major media composition	8
6.2	Table II and Table III: Online and offline outlet ideology	8
6.3	Table IV and Figure II: Main isolation-index comparison	9
6.4	Figure III and Table V: Distribution of online conservative exposure	10
6.5	Figure IV and Figure V: Time trend and concentration of online consumption	11
6.6	Table VI and Figure VI: Site size and the gap between extreme sites and extreme users	12
6.7	Table VII and Table VIII: Cross-visiting and robustness	13
6.8	Figure VII and Figure VIII: Story-level concern and demographic comparison	14
6.9	Appendix Figure A.1: Imputing the ideology of moderates	15
7	Short summary	16
8	Conclusion	17
8.1	Core conclusion	17
8.2	Economic intuition	17
8.3	Incremental contribution revisited	17

8.4 Implications for media policy and polarization research 17

8.5 Limitations 17

8.6 Takeaway 18

1 Big picture

- **Question.** Is the Internet creating ideological echo chambers, and how segregated is online news consumption relative to offline news and face-to-face interactions?
- **Motivation.** A common concern is that the Internet lowers search costs and expands choice, allowing liberals and conservatives to consume only like-minded news. The paper tests whether this concern is quantitatively important.
- **Core measure.** The paper adapts the isolation index from the racial segregation literature. In this context, ideological segregation equals the average conservative exposure of conservatives minus the average conservative exposure of liberals.
- **Main empirical setting.** The authors combine comScore data on online news audiences, MRI data on offline media consumption, and GSS/CNES data on face-to-face social networks.
- **Main result 1: online segregation is not zero, but it is low.** The Internet isolation index is 7.5 percentage points. Conservatives' average conservative exposure is 60.6%, while liberals' average conservative exposure is 53.1%.
- **Main result 2: online segregation is higher than most offline news media but lower than social interaction.** The Internet is more segregated than broadcast news, cable news, magazines, and local newspapers, but less segregated than national newspapers and far less segregated than workplaces, neighborhoods, families, trusted friends, or political discussants.
- **Main result 3: extreme online news diets are rare.** A person who reads only `nytimes.com` is more liberal than roughly 95% of online news users, while a person who reads only `foxnews.com` is more conservative than roughly 99% of online news users.
- **Mechanism 1: traffic concentration.** Most online news consumption is concentrated on large, relatively centrist sites such as Yahoo! News, AOL News, MSNBC, and CNN. Highly ideological sites exist, but they account for a small share of total visits.
- **Mechanism 2: cross-visiting.** Visitors to extreme sites often also visit mainstream or opposite-side sites. Extreme sites are more polarized than extreme users.
- **Economic takeaway.** The paper challenges a strong echo-chamber view. The Internet may allow ideological sorting, but measured online news consumption in the sample is much less segregated than many important offline social interactions.

2 Incremental contribution of the paper

- **From theoretical concern to direct measurement.** Prior debates emphasized the possibility of online echo chambers. This paper gives a direct empirical measure of ideological segregation in actual news consumption.
- **A common metric across domains.** The paper uses the same isolation-index logic for online media, offline media, geography, and face-to-face networks. This makes comparisons across domains economically interpretable.

- **Audience-based rather than content- or link-based evidence.** Much earlier evidence on online ideology uses web links, blog networks, or content. This paper uses the ideological composition of actual visitors and consumers.
- **Online-offline comparison.** The paper does not ask whether the Internet is perfectly balanced in isolation. It asks whether online news is more segregated than the realistic alternatives people might substitute away from: offline media and social interaction.
- **Distributional contribution.** The paper goes beyond an average index and studies the distribution of conservative exposure across individual Internet users, showing that highly homogeneous news diets are rare.
- **Mechanism contribution.** The paper explains why online segregation is limited: consumption is concentrated on large centrist sites, and users visit multiple outlets.
- **Robustness contribution.** The paper tests alternative assumptions about moderates, alternative ideology measures, alternative time aggregation, larger website samples, and the possibility of within-outlet story-level segregation.
- **Policy contribution.** The results discipline media-policy concerns about fragmentation. The Internet may increase choice, but the relevant comparison is not an idealized public square; it is the more segregated structure of many offline social networks.

3 Data and measurement

3.1 Online news data

- The online news data come primarily from comScore.
- The authors build a national political news and opinion website universe from comScore categories, major newspaper sites, and external lists of political websites.
- The final universe contains 1,379 sites. comScore Media Metrix size data are available for 459 sites, and comScore Plan Metrix ideology data are available for 119 large sites. These 119 sites form the main Internet sample.
- Site size is measured by average daily unique visitors in 2009.
- Site ideology is measured by the share of visitors who report being conservative among visitors who report being either conservative or liberal.
- The paper also uses comScore micro data for 2004, 2006, 2007, and 2008. The micro data allow individual-level exposure calculations and time-series analysis, but they have limitations because they are domain-level and machine-level rather than person-by-person in the same way as the aggregate data.

Interpretation: The main online measure is audience composition, not editorial slant. A site is conservative in the data if its visitors are disproportionately conservative.

3.2 Offline media data

- Offline media consumption is measured with Mediamark Research and Intelligence (MRI) Survey of the American Consumer data.

- The sample contains 51,354 respondents from the spring 2007 and spring 2008 waves.
- The paper measures consumption of cable television news, broadcast television news, national newspapers, local newspapers, and magazines.
- The MRI ideology question is directly comparable to the comScore Plan Metrix question: respondents classify themselves as conservative, liberal, or middle of the road.
- Local newspapers are defined geographically by a newspaper market, using PMSAs or counties outside PMSAs.

Interpretation: MRI makes the offline comparison possible because it provides both media consumption and political outlook for the same individuals.

3.3 Face-to-face interaction data

- The paper uses the 2006 General Social Survey (GSS) to measure the ideology of family members, neighbors, co-workers, voluntary-association contacts, and trusted people.
- The paper uses the 1992 Cross-National Election Study (CNES) to measure political discussants.
- These data allow the authors to compare media segregation with social-network segregation.
- The face-to-face measures are not media-consumption measures. They capture the ideological mix of people with whom respondents are likely to interact.

Interpretation: The benchmark is important. Even if online news is not perfectly mixed, many everyday social networks are much more ideologically homogeneous.

3.4 Share conservative and conservative exposure

For outlet j , define

$$\text{Share conservative}_j = \frac{\text{conservative visits to } j}{\text{conservative visits to } j + \text{liberal visits to } j}.$$

For individual i , conservative exposure is the average share conservative of the outlets the individual visits. If a person visits only `cnn.com`, her exposure is the share conservative of `cnn.com`. If she visits both `cnn.com` and `foxnews.com`, her exposure is the average of the two sites' conservative shares, weighted by visit frequency in the implementation.

Interpretation: Exposure converts a consumption bundle into an ideological environment. It asks: how conservative are the other users of the outlets this person consumes?

3.5 Isolation index

Let m index a medium, $j \in J_m$ index outlets in that medium, and let $cons_j$, lib_j , and $visits_j$ be conservative, liberal, and total visits to outlet j . Let $cons_m$ and lib_m be total conservative and liberal visits in medium m . The isolation index is

$$S_m = \sum_{j \in J_m} \left(\frac{cons_j}{cons_m} \cdot \frac{cons_j}{visits_j} \right) - \sum_{j \in J_m} \left(\frac{lib_j}{lib_m} \cdot \frac{cons_j}{visits_j} \right).$$

- The first term is conservatives' average conservative exposure.

- The second term is liberals’ average conservative exposure.
- If conservatives and liberals visit exactly the same outlets in the same proportions, $S_m = 0$.
- If conservatives only visit completely conservative outlets and liberals only visit completely liberal outlets, $S_m = 1$.

Interpretation: The index is not the share of conservative content. It is a difference in exposure between conservative and liberal consumers.

3.6 Moderates and ideology imputation

- The main calculation treats ideology as binary: conservative or liberal.
- Respondents who answer “middle of the road” are not discarded from site traffic. Instead, the paper imputes the share conservative among moderates visiting a site to equal the share conservative among site visitors who declare an ideology.
- Appendix Figure A.1 checks the plausibility of this assumption using ANES data on state-level moderate leaners.
- Table VIII reports robustness to alternative treatments of moderates.

Interpretation: The moderate-imputation assumption matters for levels, but the Internet isolation result remains small under many alternative assumptions.

4 Models and empirical specifications

4.1 Main comparison across media

The paper estimates an isolation index for each medium:

- online news,
- broadcast news,
- cable news,
- magazines,
- local newspapers,
- national newspapers,
- geographic units such as counties and ZIP codes,
- face-to-face networks such as work, neighborhoods, family, trusted people, and political discussants.

Interpretation: The empirical question is comparative. The Internet may be more segregated than some offline media but still less segregated than everyday social contacts.

4.2 Leave-out estimation

In finite samples, a person’s own ideology mechanically affects the measured conservative share of an outlet she visits. The appendix therefore uses a leave-out estimator when computing exposure: the outlet share used for respondent i excludes respondent i .

Interpretation: This correction prevents small-sample mechanical correlation from inflating the isolation index.

4.3 Distribution of exposure

Using comScore micro data, the authors compute individual-level conservative exposure. This lets them study the whole distribution, not only the average gap between conservatives and liberals.

Interpretation: This is crucial because a low average isolation index could hide a small group of extreme users. Figure III shows that such extreme diets exist but are uncommon.

4.4 Time-series specification

The paper uses comScore micro data to estimate online segregation from 2004 to 2008. Because the aggregate ideology data are not available for all years, the time-series estimates are scaled relative to the 2008 level.

Interpretation: The exercise is not as precise as the 2009 aggregate analysis, but it directly tests the claim that online segregation is rapidly rising over time.

4.5 Mechanism analysis

The paper studies two mechanisms that reduce segregation:

- **Site-size concentration.** Large sites dominate total visits and tend to be less ideologically extreme.
- **Cross-visiting.** Users of ideologically extreme sites often visit mainstream and opposite-side sites in the same month.

Interpretation: The Internet can contain many extreme sites without producing extreme user-level diets, because most people consume large general-interest sites or consume more than one outlet.

4.6 Robustness and story-level checks

- The paper re-estimates Internet segregation after excluding AOL and Yahoo!, adding Google News, expanding the website set, changing the treatment of moderates, and changing the ideology measure.
- It also compares daily segregation around two major news events: the 2008 U.S. presidential election and the 2007 Virginia Tech shooting.
- If liberals and conservatives normally read different stories within the same outlet, event days should increase outlet-level overlap in story consumption and potentially raise measured outlet-level segregation. The paper does not find this.

Interpretation: Robustness checks support the central result: online news segregation is not large enough to match a strong echo-chamber narrative.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies descriptive differences in ideological exposure across media and interaction domains.
- It does not estimate a randomized causal effect of the Internet on polarization or beliefs.
- It measures consumption patterns and exposure patterns, not persuasion or belief updating.

Interpretation: The correct reading is comparative measurement. The paper says online news consumption is not highly segregated relative to offline alternatives; it does not say the Internet has no political effects.

5.2 Why the comparison is informative

- A concern about the Internet is meaningful only relative to the alternatives that people would otherwise consume.
- Offline social networks are not ideologically random. Families, neighborhoods, workplaces, and political discussants are highly segregated.
- The use of a common isolation index makes the online-offline comparison transparent.

Interpretation: The Internet can be imperfectly integrated and still be less segregated than many real-world environments.

5.3 Main threats and limitations

- **Outlet-level versus story-level exposure.** The paper observes websites, newspapers, and programs, not every story read. The event-day analysis addresses but does not fully eliminate this concern.
- **Self-reported ideology.** Conservative and liberal labels come from survey self-reports. Table VIII shows similar results using party affiliation, but ideological labels remain imperfect.
- **Moderate imputation.** The main results rely on assumptions about moderates, though robustness checks show the qualitative message is stable.
- **External validity over time.** The main online sample is centered on 2009. The Internet may evolve, and later platforms or social media could have different segregation patterns.
- **Belief interpretation.** Seeing similar outlets does not guarantee that conservatives and liberals interpret the same information similarly.

Interpretation: The paper is strong evidence about exposure and consumption in its data period. It is weaker evidence about belief formation, persuasion, and the long-run political effects of the Internet.

6 Tables and figures

6.1 Figure I and Table I: Measurement validation and major media composition

Read:

- Figure I compares estimated share conservative for major online outlets using comScore and MRI. Most points lie near the 45-degree line, indicating that measured and self-reported media consumption generate similar ideological composition estimates.
- Table I shows that the U.S. adult population is 42% conservative, 21% liberal, and 38% moderate in MRI.
- The Internet and magazines have relatively more liberal visitors than the U.S. adult population, while cable and local newspapers are somewhat more conservative.
- The Internet accounts for a smaller share of daily news visits than cable, local newspapers, and broadcast news in this sample.

Economic message: Figure I supports comparability between online and offline data sources. Table I shows that online news is not just a mirror of the whole population; its audience is somewhat more liberal, but not overwhelmingly so.

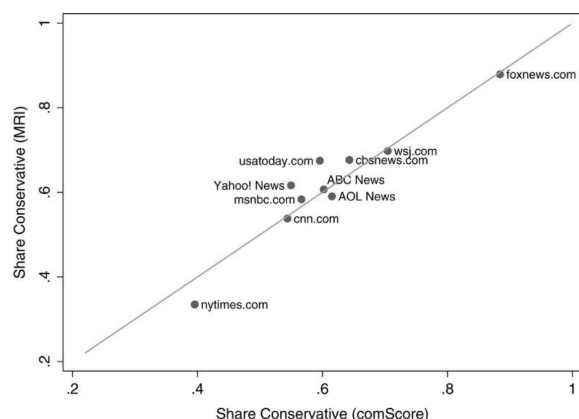


FIGURE I
Comparison of MRI and comScore Share Conservative

Data are from comScore and MRI. Share conservative is the estimated share of daily visitors who are conservative among those who report being either conservative or liberal. The line shown is the 45-degree line.

Figure I: MRI versus comScore share conservative

SIZE AND IDEOLOGICAL COMPOSITION OF MAJOR NEWS MEDIA				
U.S. adult population: 42% conservative, 21% liberal, 38% moderate				
Medium	Share of daily visitors who are:			Share of daily visits
	Conservative	Liberal	Moderate	
Cable	.45	.19	.36	.29
Local newspapers	.43	.19	.38	.29
Broadcast news	.42	.20	.38	.24
Internet	.37	.28	.35	.10
Magazines	.37	.28	.35	.05
National newspapers	.40	.31	.29	.03

Notes: Share of daily visits is the ratio of the sum of average daily unique visitors across all outlets in the medium to the sum of average daily unique visitors across all outlets in all media. Share of daily visitors who are [conservative/liberal/moderate] is the average across outlets of the share of daily visitors who report a given ideology, weighting each outlet in the medium by its average daily unique visitors. Conservative includes respondents who report that they are somewhat or very conservative; liberal includes respondents who report that they are somewhat or very liberal; moderate includes respondents who report that they are "middle of the road." Internet data are from comScore; data on other media are from MRI.

Table I: Size and ideological composition of major news media

6.2 Table II and Table III: Online and offline outlet ideology

Read:

- Table II reports the 10 largest online outlets and the most conservative and most liberal online sites.
- Large sites such as Yahoo! News, AOL News, MSNBC, CNN, and USA Today have relatively centrist audiences.
- The most ideological online sites are much more extreme: for example, several conservative host sites and liberal blogs have very lopsided conservative/liberal visitor shares.

- Table III reports offline media. Fox News is more conservative than CNN or MSNBC; the New York Times, BBC, PBS, the Atlantic, and the New Yorker have more liberal audiences.
- Offline outlets can be ideologically distinct, but the most extreme online sites are more polarized than the listed offline outlets.

Economic message: The Internet has a long tail of highly ideological sites, but most online traffic goes to large centrist sites. Therefore site extremity is not the same as user-level segregation.

TABLE II
SIZE AND IDEOLOGICAL COMPOSITION OF ONLINE NEWS OUTLETS

Ten largest				
Site	Share of daily visitors who are:			Daily UV ('000)
	Conservative	Liberal	Moderate	
drudgereport.com	.78	.06	.16	475
foxnews.com	.76	.10	.14	1,159
AOL News	.37	.23	.40	3,971
usatoday.com	.37	.25	.37	518
msnbc.com	.34	.26	.40	3,264
Yahoo! News	.31	.25	.43	6,455
cnn.com	.33	.27	.40	2,650
nytimes.com	.30	.45	.25	879
huffingtonpost.com	.22	.52	.26	583
BBC News	.16	.57	.26	472
Most conservative				
billioreilly.com	.99	.00	.01	10
rushlimbaugh.com	.97	.01	.03	43
glennbeck.com	.89	.01	.09	38
humanevents.com	.91	.03	.06	21
townhall.com	.89	.04	.08	42
thestate.com	.58	.04	.38	36
aclj.org	.85	.06	.09	18
cnsnews.com	.92	.06	.01	12
drudgereport.com	.78	.06	.16	475
realclearpolitics.com	.87	.07	.06	41
Most liberal				
thinkprogress.org	.05	.83	.12	12
blogcritics.org	.12	.61	.27	17
bvblackspin.com	.09	.43	.48	57
moveon.org	.14	.58	.28	21
BBC News	.16	.57	.26	472
blogtalkradio.com	.17	.58	.25	33
reddit.com	.15	.52	.33	36
newsvine.com	.21	.63	.16	56
alternet.org	.24	.67	.10	16
dailykos.com	.25	.68	.06	26

Notes: Average daily unique visitors is reported in 1000s. Data are from comScore. Conservative includes respondents who report that they are somewhat or very conservative; liberal includes respondents who report that they are somewhat or very liberal; moderate includes respondents who report that they are "middle of the road." "Most conservative" sites are those with the highest ratio of conservative to liberal daily visitors; "most liberal" sites are those with the highest ratio of liberal to conservative daily visitors. Sites are presented in descending order by the ratio of conservative to liberal daily visitors. To improve precision, sites with fewer than 10,000 average daily unique visitors are excluded from "most conservative" and "most liberal" lists.

Table II: Online news outlets

TABLE III
SIZE AND IDEOLOGICAL COMPOSITION OF OFFLINE NEWS OUTLETS

Magazines				
	Share of daily readers who are:			Market share
	Conservative	Liberal	Moderate	
<i>Barron's</i>	.43	.19	.37	.02
<i>U.S. News & World Report</i>	.43	.20	.37	.14
<i>BusinessWeek</i>	.42	.21	.37	.07
<i>Forbes</i>	.40	.22	.37	.04
<i>Fortune</i>	.37	.24	.39	.03
<i>TIME</i>	.35	.27	.38	.31
<i>Newsweek</i>	.37	.29	.34	.27
<i>The Economist</i>	.35	.41	.23	.04
<i>The Atlantic</i>	.24	.55	.21	.01
<i>New Yorker</i>	.17	.60	.24	.07
National newspapers				
<i>USA Today</i>	.45	.22	.33	.40
<i>Wall Street Journal</i>	.45	.21	.34	.29
<i>New York Times</i>	.26	.54	.21	.31
Broadcast news				
CBS	.42	.18	.40	.28
NBC	.44	.20	.36	.29
ABC	.42	.19	.40	.31
BBC	.37	.30	.33	.06
PBS	.32	.37	.30	.07
Cable				
Fox News	.54	.13	.33	.36
Bloomberg Television	.50	.18	.32	.01
CNBC	.41	.22	.37	.13
CNN	.40	.22	.38	.33
MSNBC	.39	.24	.36	.17

Notes: Data are from MRI. Conservative includes respondents who report that they are somewhat or very conservative; liberal includes respondents who report that they are somewhat or very liberal; moderate includes respondents who report that they are "middle of the road." Outlets are presented in descending order by the ratio of conservative to liberal daily readers/viewers. Market share is the ratio of the outlet's daily readers/viewers to the sum of daily readers/viewers across all listed outlets in the medium. Market shares may not sum to 1 due to rounding.

Table III: Offline news outlets

6.3 Table IV and Figure II: Main isolation-index comparison

Read:

- Table IV is the core result. Internet isolation is 0.075: conservatives' exposure is 0.606 and liberals' exposure is 0.531.
- Broadcast news has the lowest isolation index, 0.018. Cable, magazines, and local newspapers are also below the Internet.
- National newspapers have an isolation index of 0.104, higher than the Internet.
- Face-to-face interactions are much more segregated: voluntary associations 0.145, work 0.168, neighborhood 0.187, family 0.243, trusted people 0.303, and political discussants 0.394.

- Figure II visualizes the same ranking and makes the contrast with social networks especially clear.

Economic message: The Internet is not the most integrated news medium, but it is far from the most segregated environment in daily life. The strongest ideological sorting is in social interaction, not online news consumption.

TABLE IV
IDEOLOGICAL SEGREGATION BY MEDIUM AND TYPE OF INTERACTION

	Conservative exposure of Conservatives	Liberals	Isolation index
Internet	.606	.531	.075
Offline media			
Broadcast news	.677	.660	.018
Cable	.712	.679	.033
Magazines	.587	.540	.047
Local newspapers	.695	.647	.048
National newspapers	.612	.508	.104
Face-to-face interactions			
County	.682	.622	.059
ZIP code	.637	.543	.094
Voluntary associations	.625	.480	.145
Work	.596	.428	.168
Neighborhood	.627	.439	.187
Family	.690	.447	.243
People you trust	.675	.372	.303
Political discussants	.796	.402	.394

Notes: Internet data are from comScore. County, ZIP code, and offline media data are from MRI. Voluntary associations, work, neighborhood, family, and “people you trust” data are from the GSS. Political discussants data are from the CNES. See Section III for details on the construction of exposure and isolation measures.

Table IV: Ideological segregation by medium and interaction type

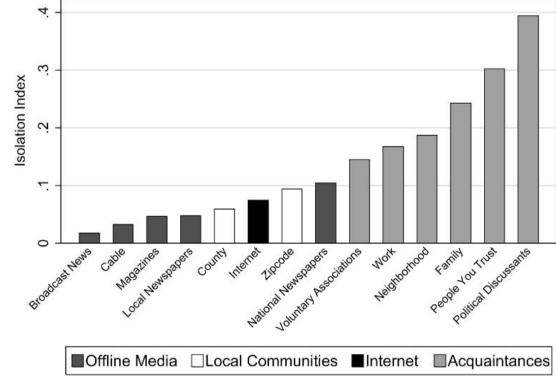


FIGURE II
Ideological Segregation by Medium and Type of Interaction
Internet data are from comScore. County, ZIP code and offline media data are from MRI. Voluntary associations, work, neighborhood, family, and “people you trust” data are from the GSS. Political discussants data are from the CNES. See Section III for details on the construction of the isolation index.

Figure II: Isolation-index ranking

6.4 Figure III and Table V: Distribution of online conservative exposure

Read:

- Figure III shows that most Internet news users have conservative exposure near the middle of the distribution.
- Half of users have exposure between about 51% and 61% conservative; the fifth percentile is about 40% and the ninety-fifth percentile is about 76%.
- Table V breaks exposure down by detailed ideology. Even very liberal users have nontrivial exposure to conservative users, and very conservative users have nontrivial exposure to liberal users.
- The “middle of the road” column is large for all ideology groups, reflecting the role of moderate audiences at many outlets.

Economic message: Echo chambers are not typical. The distribution is centered, and cross-ideology exposure remains common even for individuals with strong ideological labels.

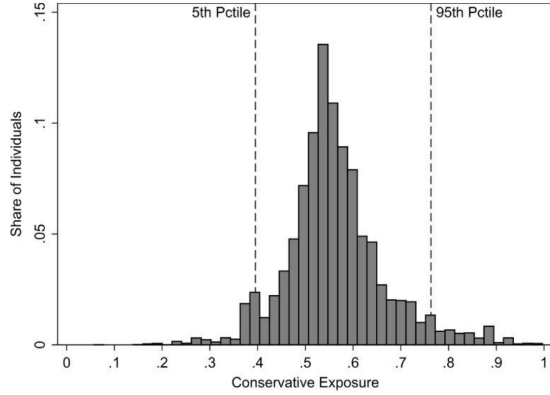


FIGURE III

Distribution of Conservative Exposure across Internet Users

Data are from comScore. An individual's conservative exposure is defined as the average share conservative on sites she visited during 2008, weighting each site by the number of days in the year on which she made at least one visit to the site. See Section III for further details on the construction of the exposure index.

Figure III: Conservative exposure across Internet users

TABLE V
EXPOSURE BY DETAILED IDEOLOGY

Exposure of:	Very liberal	Somewhat liberal	Exposure to: Middle of the road	Somewhat conservative	Very conservative
Very liberal	0.130	0.186	0.345	0.192	0.148
Somewhat liberal	0.112	0.190	0.357	0.191	0.150
Middle of the road	0.100	0.172	0.377	0.199	0.152
Somewhat conservative	0.097	0.161	0.347	0.214	0.182
Very conservative	0.087	0.147	0.309	0.212	0.246
All Internet users	0.102	0.170	0.352	0.202	0.174

Notes: Data are from comScore. See Section III for definition of exposure.

Table V: Exposure by detailed ideology

6.5 Figure IV and Figure V: Time trend and concentration of online consumption

Read:

- Figure IV shows no evidence that online ideological segregation increased from 2004 to 2008 in the comScore micro data. If anything, the index declines by 2008.
- Figure V shows that online news consumption is highly concentrated. The top four sites account for more than 50% of daily unique visits, the top 10 for more than 60%, and the top 20 for nearly 80%.
- Many small sites may be ideological, but most visits are not to those sites.

Economic message: Growth in the number of niche political websites does not mechanically imply high segregation. Market concentration in broadly appealing sites pulls aggregate exposure toward the center.

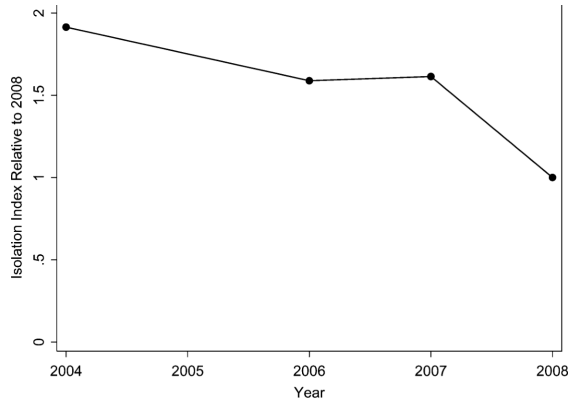


FIGURE IV

Changes in Isolation Over Time

Data are from comScore micro data. The isolation index is scaled relative to the year 2008, so that the value for 2008 is 1.00.

Figure IV: Changes in isolation over time

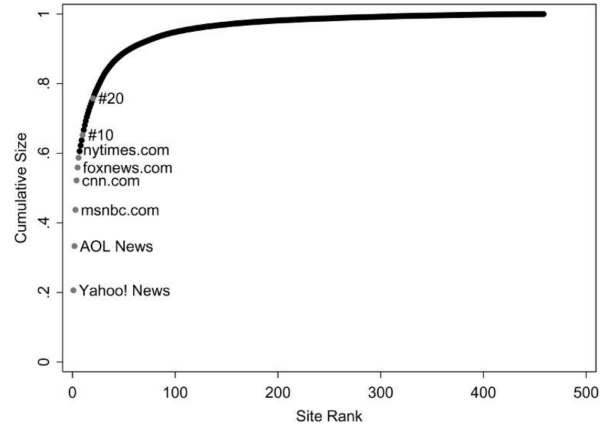


FIGURE V

Cumulative Distribution of Internet Unique Visits

Data are from comScore. Size is measured by average daily unique visitors.

Figure V: Concentration of Internet unique visits

6.6 Table VI and Figure VI: Site size and the gap between extreme sites and extreme users

Read:

- Table VI shows that the largest sites are much less segregated than the smallest sites. Sites ranked 51 and below have an isolation index of 0.213, compared with 0.062 for the top 10.
- Figure VI plots each site's share conservative against the conservative exposure of that site's average visitor.
- The regression line is much flatter than the 45-degree line. This means users of extreme sites usually also visit less extreme sites.
- Extreme sites are therefore much more common than extreme news diets.

Economic message: The long tail can look polarized at the site level, but multi-homing reduces user-level polarization. The relevant unit for democratic exposure is often the user's full diet, not the ideology of a single site.

TABLE VI
IDEOLOGICAL SEGREGATION BY SITE SIZE

Subset of sites with size rank	Share of daily visitors	Cons. exposure of		Isolation index
		Conservatives	Liberals	
1–10	.687	.599	.536	.062
11–25	.147	.584	.526	.058
26–50	.094	.610	.525	.086
51+	.065	.695	.482	.213

Notes: Data are from comScore. Share of daily visitors is the fraction of total daily unique visitors across all sites accounted for by sites in the given size group. See Section III for details on the construction of exposure and isolation measures.

Table VI: Ideological segregation by site size

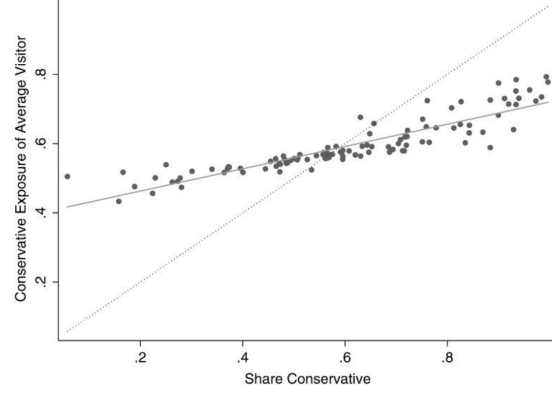


FIGURE VI

Visitor Exposure vs. Site Share Conservative

Data are from comScore. Figure plots conservative exposure of average daily visitor against the share of daily visitors who are conservative. An individual's conservative exposure is defined as the average share conservative on sites she visited during 2008, weighted by the number of days in the year on which she made at least one visit. The solid line is an OLS regression fit; the dotted line is the 45-degree line. See Section III for further details on the construction of the exposure index.

Figure VI: Visitor exposure versus site share conservative

6.7 Table VII and Table VIII: Cross-visiting and robustness

Read:

- Table VII shows that visitors to very conservative sites often visit Yahoo! News and sometimes `nytimes.com`; visitors to very liberal sites often visit Yahoo! News and sometimes `foxnews.com`.
- This cross-visiting pattern is direct evidence that niche-site users do not necessarily live in isolated information environments.
- Table VIII shows that Internet isolation remains modest across many robustness checks.
- Excluding AOL and Yahoo! raises the index to 0.113, but it remains well below many face-to-face interaction measures.
- Adding Google News, using a larger website set, changing moderate assumptions, and using party affiliation do not overturn the main conclusion.

Economic message: The low Internet isolation result is not an artifact of one portal, one moderate-imputation rule, or one ideology definition. Cross-visiting and robustness checks both support the same interpretation.

TABLE VII
CROSS-VISITING ONLINE

Share visiting in the same month			
Site	Yahoo! News	foxnews.com	nytimes.com
Monthly visitors of:			
Any Internet site	.24	.05	.06
Yahoo! News	1.00	.09	.12
Most conservative			
billoreilly.com	.38	.50	.22
rushlimbaugh.com	.50	.49	.31
glennbeck.com	.44	.44	.21
humanevents.com	.51	.44	.34
townhall.com	.51	.42	.33
thetate.com	.43	.28	.21
aclj.org	.42	.25	.15
cnsnews.com	.61	.60	.44
drudgereport.com	.52	.44	.30
realclearpolitics.com	.60	.53	.51
Most liberal			
thinkprogress.org	.57	.33	.48
blogcritics.org	.30	.13	.21
bvblackspin.com	.25	.12	.14
moveon.org	.41	.12	.27
BBC News	.39	.18	.25
blogtalkradio.com	.24	.07	.14
reddit.com	.35	.12	.28
newsvine.com	.37	.24	.21
alternet.org	.45	.24	.40
dailykos.com	.45	.24	.40

Notes: The table reports the share of all monthly unique visitors to a given site (listed in the first column) that make at least one visit in the same month to Yahoo! News, nytimes.com, and foxnews.com. These data are taken from comScore Media Metrix and are averaged over the 12 months of 2009.

Table VII: Cross-visiting online

TABLE VIII
ROBUSTNESS CHECKS: INTERNET SEGREGATION

	Conservative exposure of		
	Conservatives	Liberals	Isolation index
Baseline	.606	.531	.075
Exclude AOL & Yahoo!	.622	.509	.113
Expand the set of news sites			
Add Google News	.601	.530	.072
391 Websites in			
comScore micro data	.616	.517	.099
All news websites upper			
bound	.617	.516	.102
Moderates			
Treat as conservatives	.742	.692	.050
Treat as liberals	.425	.346	.079
Drop	.618	.528	.091
Treat as drawn at random	.598	.551	.047
Treat as 50-50	.574	.525	.048
Alternative ideology measures			
Political party	.522	.451	.071
Right-of-median ZIP code	.510	.497	.013

Notes: Data are from comScore. See Section III for details on the construction of exposure and isolation measures. ZIP code ideology measure is constructed from Federal Election Commission data on political contributions. See Section VI for details.

Table VIII: Robustness checks

6.8 Figure VII and Figure VIII: Story-level concern and demographic comparison

Read:

- Figure VII studies the 2008 U.S. election and the 2007 Virginia Tech shooting. News visits spike on major event days, but the daily isolation index does not spike.
- This weakens the concern that outlet-level measures hide large story-level segregation, at least for these major common-interest events.
- Figure VIII compares ideological-style measurement with demographic segregation by gender, race, income, and education.
- Racial and educational segregation are far larger in geography and political discussion networks than in most news media.
- Gender segregation in media is low, though the Internet is the most segregated listed news medium for gender.

Economic message: Event-day behavior suggests that liberals and conservatives do not split into completely different sources for major news. The demographic comparison also places Internet ideology in perspective: U.S. society is far more segregated along several social-network dimensions.

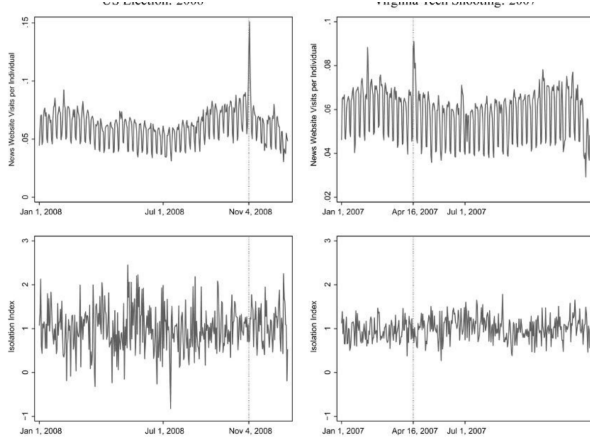


FIGURE VII

Online Daily Visitors and Segregation by Day

Data are from comScore micro data. In top panel, news website visits per individual is the average across individuals of the number of news websites in our main sample visited on each day. In bottom panel, the isolation index is scaled so that its mean across days is equal to 1.

Figure VII: Daily visitors and segregation around major news events

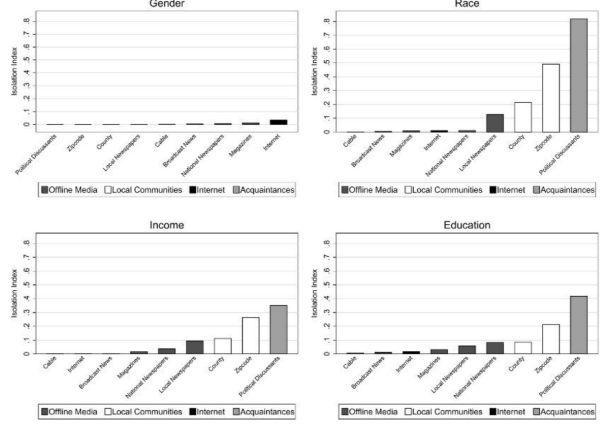


FIGURE VIII

Demographic Segregation by Medium and Type of Interaction

For each demographic characteristic we divide respondents into two groups to compute the isolation index defined in Section III. To compute gender isolation, we divide respondents into males and females according to the gender of the respondent in comScore PlanMetrix (Internet), CNES (Political Discussants), and MRI (geography and other media). To compute racial isolation, we divide respondents into blacks and whites (excluding all others) according to the race of the respondent in 2008 comScore micro data (Internet), CNES (Political Discussants), and MRI (geography and other media). To compute income isolation, we divide respondents into those with above- and below-median income according to the household income of the respondent in 2008 comScore micro data (Internet), CNES (Political Discussants), and MRI data (geography and other media). To compute education isolation, we divide respondents into those who graduated from a 4 year college and those who did not according to the highest education level in the household in 2002 comScore micro data (Internet) and MRI data (geography other media), and the education of the respondent in CNES (Political Discussants). We truncate negative values of the isolation index at 0.

Figure VIII: Demographic segregation by medium and interaction type

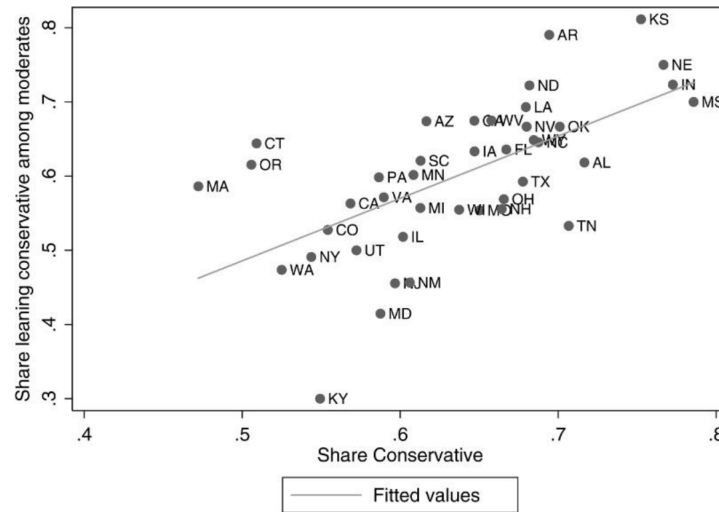
6.9 Appendix Figure A.1: Imputing the ideology of moderates

Read:

- Appendix Figure A.1 uses ANES state-level data to compare the conservative share among people who declare an ideology with the conservative-leaning share among moderates.
- The fitted relationship is positive and close to the assumption used in the main imputation.
- The paper reports that the estimated slope is statistically different from zero but not statistically different from one.

Economic message: The figure supports the idea that moderates' leanings track the ideological environment around them. This makes the paper's baseline imputation more plausible, although it does not directly test outlet-level moderate behavior.

you thought much about this:



APPENDIX FIGURE A.1

Imputing the Ideology of Moderates

Data are from the American National Election Study (www.electionstudies.org), years 1988, 1992, 1996, 1998, 2000, 2002, 2004, 2008. The unit of observation is the U.S. state. The x -axis shows the fraction conservative among those declaring an ideology. The y -axis shows the fraction conservative among moderates who report a “leaning.” Sample excludes states with fewer than 25 moderate “leaners” during the sample period.

Appendix Figure A.1: Imputing the ideology of moderates

7 Short summary

- The paper studies ideological segregation in online news, offline news, geography, and face-to-face interaction.
- It uses a common isolation-index measure from the segregation literature.
- Online news segregation is positive but modest: the Internet isolation index is 7.5 percentage points.
- Online news is more segregated than broadcast news, cable, magazines, and local newspapers, but less segregated than national newspapers.
- Face-to-face social networks are much more segregated than online news.
- Extremely ideological online sites exist, but they are small relative to large centrist news portals.
- Most users have moderate news diets; few resemble a pure `foxnews.com`-only or `nytimes.com`-only consumer.
- Users of extreme sites often cross-visit mainstream and opposite-side sites.
- There is no evidence in the sample that Internet news became more segregated from 2004 to 2008.

- The main caveat is that exposure is not the same as belief formation: people may interpret the same information differently.

8 Conclusion

8.1 Core conclusion

Gentzkow and Shapiro show that online news consumption is not characterized by extreme ideological segregation in their data. The Internet isolation index is low in absolute terms and much lower than the segregation of many face-to-face interactions. The result does not imply that online news is perfectly balanced, but it substantially weakens the strongest echo-chamber narrative for the period studied.

8.2 Economic intuition

The result follows from two simple economic forces. First, attention is scarce and concentrated: most users go to a few large, general-interest news sites. Second, consumers can multi-home: visiting one ideological site does not prevent them from also visiting mainstream or opposing-side sites. These forces mean that a world with many ideological websites need not produce highly ideological individual news diets.

8.3 Incremental contribution revisited

The paper advances the media and political-economy literature by moving from speculation about fragmentation to direct measurement of audience overlap. Its most important conceptual step is to use the same segregation metric for online media, offline media, geography, and social networks. This makes the finding meaningful: online segregation is not evaluated against a utopian benchmark, but against realistic offline alternatives.

8.4 Implications for media policy and polarization research

The evidence suggests that worries about polarization should not focus only on online news outlet choice. Offline social networks, families, neighborhoods, and political discussion partners are often much more segregated. Policies or theories that treat the Internet as uniquely isolating may therefore miss larger sources of homogeneous exposure. At the same time, the Internet may still matter through mechanisms not measured here, such as algorithmic ranking, social media sharing, selective interpretation, or differential trust.

8.5 Limitations

The paper measures exposure to outlets, not direct exposure to every article, headline, or conversation. The data mainly cover the late 2000s and predate the later dominance of several social-media platforms. The ideology measures rely on self-reports and imputation for moderates. The analysis is not a causal estimate of the Internet's effect on polarization. Finally, the paper explicitly does not measure how people transform encountered information into beliefs.

8.6 Takeaway

The enduring takeaway is that ideological segregation online is real but limited in this setting. The Internet contains ideological niches, but typical users' news diets are more mixed than the strongest echo-chamber claims suggest. For understanding polarization, the paper points researchers toward the interaction between media exposure, social networks, and belief formation rather than toward online outlet choice alone.